

Question	Answer	Marks
6(a)	a flow of charge carriers	B1
6(b)	a straight line from 0 A, 1.50 V with a negative gradient	M1
	straight line passes through 1.60 A, 0.30 V	A1
6(c)(i)	$\rho = RA / L$ $= 0.82 \times 2.0 \times 10^{-7} / 6.8$	C1
	$= 2.4 \times 10^{-8} \Omega \text{m}$	A1
6(c)(ii)	terminal pd $= 1.50 - (1.8 \times 0.75)$ $= 0.15$	C1
	current in fixed resistor $= 1.8 - (0.15 / 0.82)$ $= 1.6$	C1
	$R = 0.15 / 1.6$ $= 0.093 \Omega$	A1
	OR terminal pd $= 1.50 - (1.8 \times 0.75)$ $= 0.15$	(C1)
	total external resistance $= 0.15 / 1.8$ $= 0.083$ $1 / 0.083 = 1 / 0.82 + 1 / R$	(C1)
	$R = 0.093 \Omega$	(A1)

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6(c)(iii)	$P = IV$ or $P = V^2/R$ or $P = I^2R$ $= 1.8 \times (1.50 - 0.15)$ or $(1.50 - 0.15)^2 / 0.75$ or $1.8^2 \times 0.75$	C1
	$= 2.4 \text{ W}$	A1